

GROUNDWATER EXPLORATION

Comprehensive Engineering Module

Syllabus Coverage: Chapter 6

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Department of Civil Engineering
Tribhuvan University

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1 6.1 Objectives of Groundwater Exploration

1.1 Introduction

Groundwater exploration is the fundamental investigative process used to define the hydrogeological regime of an area. Unlike surface water hydrology, where the resource is visible and flow can be measured directly using weirs or current meters, groundwater is an occluded resource. Its presence, quantity, and quality must be inferred from indirect surface measurements or expensive point-source subsurface data.

The exploration phase serves as the critical bridge between the theoretical demand for water and the engineering design of supply systems. In the context of Nepal, where hydrogeological conditions vary drastically from the unconsolidated alluvial sediments of the Terai to the complex fractured metamorphic rocks of the Lesser Himalayas, exploration is not merely a formality but a necessity for project viability.

1.2 Detailed Objectives

The primary objective of any groundwater exploration program is to locate a source of water that is **sustainable**, **economical**, and **safe**. However, this high-level goal comprises several granular technical objectives.

1.2.1 1. Definition of Hydrogeological Boundaries (Geometry)

Before any quantitative analysis can be performed, the physical framework of the aquifer system must be established.

- **Vertical Zonation:** Determining the depth to the water table (in unconfined systems) or the piezometric surface (in confined systems). This dictates the type of pump required (suction vs. submersible) and the static lift head, which drives operational costs.
- **Stratigraphic Mapping:** Identifying the depth and thickness of the aquifer. The thickness (b) is a critical parameter because Transmissivity (T), the measure of an aquifer's ability to transmit water, is the product of Hydraulic Conductivity (K) and thickness (b). A thin layer of highly permeable sand may yield less water than a thick layer of moderately permeable sandstone.
- **Lateral Extent:** Determining if the aquifer is a localized lens (common in point-bar deposits) or a regionally extensive sheet. This impacts the long-term reliability of the source during drought years.
- **Boundary Conditions:** Identifying impermeable boundaries (barrier boundaries) such as fault scarps or clay dikes which will stop flow, versus recharge boundaries (like rivers or lakes) which will replenish the aquifer during pumping.

1.2.2 2. Estimation of Hydraulic Parameters (Quantity)

The "Quantity" objective is not just about finding water, but defining the aquifer's capacity to store and transmit it. Exploration provides initial estimates for:

- **Porosity (n):** The ratio of void volume to total volume. This defines the total water in storage.
- **Specific Yield (S_y):** The volume of water that will drain under gravity. In unconfined aquifers, this is the water actually available for extraction.
- **Hydraulic Conductivity (K):** A measure of the ease with which water moves through the pore spaces.
- **Transmissivity (T):** $T = Kb$. This is the parameter most directly related to the specific capacity of a well.
- **Storage Coefficient (S):** In confined aquifers, this represents the water released from storage per unit decline in hydraulic head due to the expansion of water and compression of the rock skeleton.

1.2.3 3. Assessment of Hydrochemistry (Quality)

Water quality is a defining constraint. A high-yield aquifer is useless if the water is chemically hazardous or too saline for use. Exploration aims to map:

- **Total Dissolved Solids (TDS):** To distinguish between fresh water, brackish water, and brine. In the deep Terai aquifers, ancient marine deposits can result in high salinity at depth. Resistivity surveys are particularly adept at identifying the freshwater-saline interface.
- **Geogenic Contaminants:** Identifying zones prone to Arsenic (common in Nawalparasi, Rautahat) or Fluoride contamination.
- **Anthropogenic Indicators:** Detecting high Nitrate levels in shallow aquifers which indicate contamination from fertilizers or sewage.

1.2.4 4. Optimization of Drilling Strategy (Economics)

Drilling is the most expensive component of groundwater development.

- **Depth Optimization:** Exploration prevents "over-drilling". If a good aquifer exists at 80m, drilling to 200m wastes money on fuel, casing, and rig time.
- **Method Selection:** Knowing the lithology determines the rig type. Percussion (Cable Tool) rigs are needed for the boulder-heavy Bhabar zone, while Rotary rigs are efficient for the sandy Terai. Attempting to use a rotary rig in a boulder zone can result in lost drill bits and stuck pipes.

1.2.5 5. Risk Mitigation

In engineering economics, the cost of exploration is viewed as an insurance premium against the risk of a dry hole.

- **The Dry Hole Cost:** A failed well costs 100% of the drilling budget with zero return.
- **The Exploration Cost:** Typically 5-10% of the drilling cost.
- **Benefit-Cost Analysis:** If an exploration program increases the success rate from 60% to 90%, the expenditure is economically justified.

Past Exam Questions

Frequently Asked Questions:

- **2073 Bhadra [2 Marks]:** Discuss briefly the purpose of groundwater exploration.
- **2072 Magh (Back) [2 Marks]:** What are the objectives of groundwater exploration?
- **2072 Ashwin [2 Marks]:** What is the main objective of groundwater exploration?
- **2071 Bhadra [1+1 Marks]:** What is Ground Water Exploration? What are the objectives of Groundwater Exploration?

2 6.2 Overview of Methods of Exploration

Methods are taxonomized based on their interaction with the subsurface environment. They range from purely observational surface techniques to invasive subsurface sampling.

2.1 A. Surface Methods (Indirect)

Surface methods rely on the measurement of physical parameters at the ground surface to interpret subsurface conditions. They are non-invasive and generally cost-effective for covering large areas.

2.1.1 1. Esoteric Methods

These methods are not based on standard scientific principles but rely on human sensitivity or traditional beliefs.

- **Water Witching (Dowsing):** The use of forked sticks, rods, or pendulums.
- **Observation of Anthills/Termite Mounds:** In some cultures, high moisture content in termite mounds is taken as an indicator of a shallow water table.

2.1.2 2. Geological Methods

These involve the direct observation of surface features to infer subsurface geology.

- **Field Mapping:** Recording rock types, strike, and dip.
- **Structural Analysis:** Mapping faults, folds, and joints.
- **Stratigraphic Correlation:** Projecting known layers across distances.

2.1.3 3. Remote Sensing and Photogeology

Using aerial or satellite data to analyze large-scale features.

- **Lineament Analysis:** Identifying straight lines on satellite images which often correspond to fracture zones or vertical faults.
- **Vegetation Indices (NDVI):** Using infrared imagery to detect healthy vegetation in dry seasons, indicating groundwater availability.
- **Drainage Morphometry:** Analyzing stream patterns (dendritic, trellis, radial) to infer rock permeability and structural control.

2.1.4 4. Geophysical Methods

The application of physics to measure earth properties.

- **Electrical Resistivity:** Measuring the earth's resistance to electric current. (Most Common).
- **Seismic Refraction:** Measuring the speed of shock waves. (Second Most Common).
- **Gravity Method:** Measuring minute variations in the earth's gravitational field caused by density differences (e.g., dense bedrock vs. loose sediment).
- **Magnetic Method:** Measuring anomalies in the earth's magnetic field, useful for detecting igneous dykes or metallic ore bodies.
- **Electromagnetic (EM) Method:** Inducing currents in the ground using magnetic fields. Useful for rapid reconnaissance.
- **Ground Penetrating Radar (GPR):** Using high-frequency radio waves. Excellent for very shallow (0-30m) investigations but fails in conductive clay soils.

2.2 B. Subsurface Methods (Direct)

These methods involve penetrating the earth to obtain "Ground Truth".

2.2.1 1. Test Drilling

- **Auger Drilling:** Suitable for shallow, soft soil investigations.
- **Wash Boring:** Using high-pressure water jets; limited to soft sediments.
- **Cable Tool (Percussion):** Lifting and dropping a heavy chisel bit. Excellent for sampling but slow.
- **Direct Rotary:** Rotating a bit while circulating drilling mud. Fast and effective for deep wells.
- **Reverse Circulation Rotary:** Water flows down the annulus and up the drill pipe. Best for large-diameter production wells in unconsolidated formations.
- **Down-the-Hole (DTH) Hammer:** Using compressed air to drive a pneumatic hammer. Standard for hard rock drilling in the Himalayas.

2.2.2 2. Borehole Geophysical Logging (E-Logging)

Once a hole is drilled, sensors are lowered to record continuous depth profiles.

- **Spontaneous Potential (SP):** Measures natural voltage. Distinguishes permeable (sand) from impermeable (clay).
- **Resistivity (Short/Long Normal):** Measures formation resistivity. Confirms fresh water vs. saline water.
- **Gamma Ray:** Measures natural radioactivity. Clays are naturally radioactive (K-40); sands are not. Useful for correlation across wells.
- **Caliper Log:** Measures hole diameter. Identifies washouts or caving zones.
- **Temperature Log:** Identifies zones where water enters the well (temperature anomalies).

Past Exam Questions

Frequently Asked Questions:

- **2072 Ashwin [3 Marks]:** List out different methods of surface investigations used in groundwater exploration.
- **2071 Bhadra [6 Marks]:** Write about methods of Groundwater exploration.
- **2078 Chaitra [4 Marks]:** What are the geophysical methods for the groundwater exploration? Write in detail of any of them.

3 6.3 Water Witching (Dowsing)

Water witching, dowsing, or divining is perhaps the most controversial topic in groundwater exploration. While dismissed by the scientific community as pseudoscience, it persists globally, particularly in rural agrarian societies, including Nepal.

3.1 Historical Context

Dowsing has been recorded since the 16th century, originally used in Germany to find silver ore (not water). It was brought to England by German miners and later adapted for finding water. The term "Water Witching" gained popularity in the United States, particularly in rural Appalachia and the West.

3.2 Methodology

The procedure is ritualistic rather than technical:

1. The Device:

- **Forked Stick:** A Y-shaped branch is cut from a green tree (Willow, Peach, or Witch Hazel are preferred for their flexibility). The dowser holds the two branches of the Y with palms upward, putting the stick under tension.
- **L-Rods:** Two L-shaped metal wires (often copper or simple coat hangers). The short leg is held in the fist, with the long leg pointing forward.
- **Pendulum:** A crystal or weight on a string.

2. **The Survey:** The dowser walks slowly over the land, concentrating on the desire for water.

3. **The Signal:**

- The forked stick snaps downward (or upward) violently.
- The L-rods cross each other.
- The pendulum begins to rotate.

3.3 The Scientific Explanation: Ideomotor Effect

Extensive double-blind testing has been conducted by physicists and geologists. The consensus is that dowsing does not detect any physical signal from the water. Instead, it relies on the **Ideomotor Effect**.

The Ideomotor Effect is a psychological phenomenon where a subject makes motions unconsciously.

1. **Subconscious Observation:** An experienced farmer or local dowser knows the land. They subconsciously notice signs: a patch of green grass, a slight depression in the topography, or a specific type of tree.
2. **Expectation:** Based on these cues, their brain forms an expectation that "water is here".
3. **Muscle Reaction:** This expectation triggers minute, involuntary muscle tremors in the arms.
4. **Amplification:** The dowsing device is held in a state of unstable equilibrium. The Y-stick is under tension; the L-rods are balanced on the fulcrum of the hand. Even a microscopic muscle movement is mechanically amplified into a large movement of the rod.
5. **Confirmation Bias:** If a well is dug and water is found, the "success" is remembered. If it is dry, it is forgotten or blamed on "bad energy". In regions like the Terai, where the water table is continuous, the probability of finding water is 100%, making the dowser appear infallible.

Past Exam Questions

Frequently Asked Questions:

- **2072 Ashwin [1 Mark]:** What do you mean by water witching?

4 6.4 Geological Methods of Exploration

Geological methods involve the qualitative analysis of the earth's surface to interpret the subsurface structure. It is the "diagnosis" phase before the "testing" phase of geophysics.

4.1 1. Lithological Controls

Understanding the rock type is the first step.

- **Unconsolidated Sediments:**
 - **Gravel:** Highest permeability. Found in Bhabar zone, ancient river channels.
 - **Sand:** Good permeability. The primary aquifer of the Terai.
 - **Silt/Clay:** Impermeable. Forms aquitards and aquicludes.
- **Sedimentary Rocks:**

- *Sandstone*: Porous and permeable.
- *Limestone*: Secondary porosity through dissolution (Karst). Can yield massive springs.
- *Shale*: Impermeable.
- **Crystalline Rocks (Igneous/Metamorphic):**
 - *Granite/Gneiss/Schist*: Inherently impermeable (Primary porosity < 1%). Groundwater exists only in fractures.

4.2 2. Stratigraphy and Dip/Strike

Stratigraphy is the study of rock layers (strata).

- **Dip**: The angle at which a rock layer goes into the ground.
- **Strike**: The compass direction of the line formed by the intersection of a rock surface with a horizontal plane.

Application: If a permeable sandstone outcrop is observed at the surface dipping at 30 degrees North, a hydrogeologist can calculate exactly how deep that aquifer will be at a proposed well site 500m North.

$$\text{Depth} = \text{Distance} \times \tan(\text{Dip Angle})$$

4.3 3. Structural Geology

In the Himalayas, rock deformation controls water occurrence.

4.3.1 Fracture Trace Analysis

In hard rock, water is stored in cracks.

- **Joints**: Regular cracking patterns in rock. Interconnected joints form an aquifer.
- **Faults**: Major breaks in the crust. The fault zone often contains "Fault Breccia" (crushed rock) which is highly permeable. However, faults can also be filled with "Fault Gouge" (clay paste) which is impermeable.

4.3.2 Folds

- **Synclines (Troughs)**: Groundwater flows under gravity towards the axis of the syncline. Confined conditions often develop here, leading to Artesian wells.
- **Anticlines (Arches)**: The crest of an anticline is usually under tension (cracked), allowing infiltration, but water flows away from the crest down the limbs. The crest itself is often a dry zone.

4.4 4. Geomorphological Indicators

- **Valley Fills**: In mountainous terrain, river valleys often contain thick deposits of sand and gravel which form excellent shallow aquifers.
- **Alluvial Fans**: Fan-shaped deposits where a river exits a mountain range (e.g., Dharan, Butwal). These are coarse-grained and highly permeable recharge zones.
- **Terraces**: Flat steps on valley sides. Often contain perched aquifers.

4.5 5. Hydrobotanical Indicators (Phreatophytes)

Plants can be indicators of groundwater depth and quality.

- **Halophytes:** Salt-tolerant plants. Their presence indicates saline groundwater.
- **Phreatophytes:** Deep-rooted plants that draw water directly from the water table.
 - *Populus (Poplar) & Salix (Willow)*: Indicate fresh, shallow water.
 - *Prosopis (Mesquite)*: Can send roots down 10-30 meters.

5 6.5 Electrical Resistivity Method

This is the standard geophysical method for groundwater exploration.

5.1 Theoretical Background

5.1.1 Ohm's Law in Geophysics

Resistance R is given by V/I . Resistivity ρ is the property of the material.

$$\rho = R \frac{A}{L}$$

In a homogeneous, isotropic half-space (the earth), if we inject current I at a point source on the surface, the current flows radially outward in hemispherical shells. The surface area of a hemisphere at distance r is $2\pi r^2$. The current density J is:

$$J = \frac{I}{2\pi r^2}$$

The electric field E is related to J by ρ :

$$E = \rho J = \rho \frac{I}{2\pi r^2}$$

Since potential gradient is the negative of the electric field ($E = -dV/dr$):

$$-\frac{dV}{dr} = \rho \frac{I}{2\pi r^2}$$

Integrating from distance r to infinity (where $V = 0$):

$$V = \int_r^\infty \rho \frac{I}{2\pi r^2} dr = \frac{\rho I}{2\pi} \left[-\frac{1}{r} \right]_r^\infty = \frac{\rho I}{2\pi r}$$

This fundamental equation $V = \frac{\rho I}{2\pi r}$ gives the potential at any distance r from a single current source.

5.1.2 Four-Electrode System

In practice, we must have a closed circuit (Source and Sink).

- Current Source ($+I$) at A .
- Current Sink ($-I$) at B .
- We measure potential difference between two points M and N .

The potential at M (V_M) is the sum of the potential caused by A and B :

$$V_M = \frac{\rho I}{2\pi} \left(\frac{1}{AM} - \frac{1}{BM} \right)$$

The potential at N (V_N) is:

$$V_N = \frac{\rho I}{2\pi} \left(\frac{1}{AN} - \frac{1}{BN} \right)$$

The measured potential difference $\Delta V = V_M - V_N$ is:

$$\Delta V = \frac{\rho I}{2\pi} \left[\left(\frac{1}{AM} - \frac{1}{BM} \right) - \left(\frac{1}{AN} - \frac{1}{BN} \right) \right]$$

Solving for ρ :

$$\rho = \frac{2\pi\Delta V}{I} \left[\frac{1}{\frac{1}{AM} - \frac{1}{BM} - \frac{1}{AN} + \frac{1}{BN}} \right]$$

The term in the large brackets is the inverse of the geometric factor.

$$\rho = K \frac{\Delta V}{I}$$

5.2 Electrode Configurations

5.2.1 1. Wenner Array

Electrodes are equally spaced at distance a .

- $AM = a$
- $MN = a$
- $NB = a$
- $BM = 2a$
- $AN = 2a$
- $BN = a$

Substitute into the general equation:

$$\frac{1}{K} = \frac{1}{2\pi} \left(\frac{1}{a} - \frac{1}{2a} - \frac{1}{2a} + \frac{1}{a} \right) = \frac{1}{2\pi} \left(\frac{2}{a} - \frac{1}{a} \right) = \frac{1}{2\pi a}$$

$$K = 2\pi a$$

Therefore, for Wenner: $\rho_a = 2\pi a R$.

5.2.2 2. Schlumberger Array

The potential electrodes (MN) are kept close together, while current electrodes (AB) are far apart. Let $AB/2 = L$ (Half current spacing). Let $MN/2 = b$ (Half potential spacing).

- $AM = L - b$
- $BM = L + b$
- $AN = L + b$
- $BN = L - b$

Substituting into general equation:

$$\frac{1}{K} = \frac{1}{2\pi} \left(\frac{1}{L-b} - \frac{1}{L+b} - \frac{1}{L+b} + \frac{1}{L-b} \right) = \frac{1}{2\pi} \left(\frac{2}{L-b} - \frac{2}{L+b} \right)$$

$$\frac{1}{K} = \frac{1}{\pi} \left(\frac{(L+b) - (L-b)}{L^2 - b^2} \right) = \frac{1}{\pi} \frac{2b}{L^2 - b^2}$$

$$K = \pi \frac{L^2 - b^2}{2b}$$

Therefore, for Schlumberger: $\rho_a = \pi \frac{(AB/2)^2 - (MN/2)^2}{MN} R$.

5.3 Vertical Electrical Sounding (VES)

5.3.1 Field Procedure

1. **Site Selection:** Choose a location away from power lines, buried pipes, or fences which cause interference.
2. **Setup:** Drive electrodes into the ground. Connect cables.
3. **Measurement 1:** Set $AB/2 = 2m$, $MN/2 = 0.5m$. Inject current, measure resistance.
4. **Expansion:** Expand current electrodes to $AB/2 = 4m, 6m, 8m, 10m$.
5. **MN Shift:** When the voltage signal becomes too weak (because AB is very large), increase the MN spacing. This creates a "overlap" point in the data curve.
6. **Continuation:** Continue expanding $AB/2$ up to 100m or 200m depending on target depth.

5.3.2 Data Interpretation

We plot ρ_a vs $AB/2$ on Log-Log paper. The curve shape tells the story.

Type Curves (3 Layer Models):

1. **H-Type (Minimum Type):** $\rho_1 > \rho_2 < \rho_3$.
 - Curve drops down and then rises.
 - *Interpretation:* High resistivity dry soil over **Low resistivity water-bearing sand** over High resistivity bedrock.
 - **Verdict:** Best target for drilling.
2. **K-Type (Maximum Type):** $\rho_1 < \rho_2 > \rho_3$.
 - Curve rises up and then drops.
 - *Interpretation:* Conductive topsoil over **Resistive hard rock** over Conductive clay base.
 - **Verdict:** Poor target (unless the hard rock is fractured limestone).
3. **A-Type (Ascending):** $\rho_1 < \rho_2 < \rho_3$.
 - Curve rises continuously.
 - *Interpretation:* Soft soil over weathered rock over hard rock.
 - **Verdict:** Common in hills. Drilling depends on thickness of weathered zone.
4. **Q-Type (Descending):** $\rho_1 > \rho_2 > \rho_3$.
 - Curve falls continuously.
 - *Interpretation:* Dry sand over wet sand over saline clay.
 - **Verdict:** Caution required (risk of salinity or clay).

Past Exam Questions

Frequently Asked Questions:

- **2081 Final [4 Marks]:** Explain the electric resistivity method of sub surface exploration.
- **2073 Bhadra [4 Marks]:** Explain with necessary sketches the Electrical Resistivity method of geophysical exploration.

6.6 Seismic Refraction Method

Seismic refraction utilizes the propagation of elastic waves to map subsurface velocity contrasts.

6.1 Fundamental Physics

6.1.1 Elastic Constants

Seismic velocity depends on the elastic properties of the medium.

$$V_p = \sqrt{\frac{K + \frac{4}{3}\mu}{\rho_{density}}}$$

Where:

- K = Bulk Modulus (Incompressibility)
- μ = Shear Modulus (Rigidity)
- $\rho_{density}$ = Density

Since water has no shear strength ($\mu = 0$) but high bulk modulus (K), P-waves travel through it ($V \approx 1500m/s$), but S-waves do not. This is why we focus on P-waves.

6.1.2 Snell's Law

When a wave hits an interface between two layers with velocities V_1 and V_2 (where $V_2 > V_1$), the wave refracts.

$$\frac{\sin i}{\sin r} = \frac{V_1}{V_2}$$

Critical refraction occurs when the angle of refraction $r = 90^\circ$ (wave travels along the boundary).

$$\sin i_c = \frac{V_1}{V_2}$$

6.2 Two-Layer Case Derivation

Consider a source at origin ($x = 0$) and a geophone at distance x .

6.2.1 1. Direct Wave

The wave travels along the surface through layer 1.

$$T_{direct} = \frac{x}{V_1}$$

This plots as a straight line through the origin with slope $1/V_1$.

6.2.2 2. Refracted Wave

Path: Source $\rightarrow$ Interface (down) $\rightarrow$ Interface (along) $\rightarrow$ Surface (up). Let thickness be z . Distance traveled down = $z / \cos(i_c)$. Distance traveled up = $z / \cos(i_c)$. Horizontal distance traveled in layer 2 = $x - 2z \tan(i_c)$.

Total Time ($T_{refracted}$):

$$T_{ref} = \frac{z}{V_1 \cos i_c} + \frac{x - 2z \tan i_c}{V_2} + \frac{z}{V_1 \cos i_c}$$

$$T_{ref} = \frac{2z}{V_1 \cos i_c} + \frac{x}{V_2} - \frac{2z \sin i_c}{V_2 \cos i_c}$$

Using Snell's Law ($1/V_2 = \sin i_c / V_1$):

$$T_{ref} = \frac{x}{V_2} + \frac{2z}{V_1 \cos i_c} - \frac{2z \sin^2 i_c}{V_1 \cos i_c}$$

$$T_{ref} = \frac{x}{V_2} + \frac{2z}{V_1 \cos i_c} (1 - \sin^2 i_c)$$

Since $1 - \sin^2 i_c = \cos^2 i_c$:

$$T_{ref} = \frac{x}{V_2} + \frac{2z \cos i_c}{V_1}$$

This is the equation of a straight line ($y = mx + c$).

- Slope = $1/V_2$
- Intercept (T_i) = $\frac{2z \cos i_c}{V_1}$

Using $\cos i_c = \sqrt{1 - \sin^2 i_c} = \sqrt{1 - (V_1/V_2)^2} = \frac{\sqrt{V_2^2 - V_1^2}}{V_2}$:

$$T_i = \frac{2z}{V_1} \frac{\sqrt{V_2^2 - V_1^2}}{V_2}$$

Solving for depth z :

$$z = \frac{T_i V_1 V_2}{2\sqrt{V_2^2 - V_1^2}}$$

6.3 Field Operation and Interpretation

6.3.1 Equipment

- **Seismograph:** The recording unit. Modern units are 12 or 24 channel.
- **Geophones:** Detectors spiked into the ground. For P-wave surveys, vertical component geophones (14 Hz or 28 Hz) are used.
- **Cable:** Connects geophones to the seismograph.
- **Source:** 8kg Sledgehammer and steel plate.

6.3.2 Interpretation Steps

1. **Picking First Arrivals:** On the seismogram, identify the exact moment the first vibration starts.
2. **Plotting:** Plot Time (Y) vs Distance (X).
3. **Fitting Lines:** Draw best-fit straight lines through the points.
4. **Calculating Velocities:** Invert the slopes to get V_1 and V_2 .
5. **Calculating Depth:** Extrapolate the second line to the Y-axis to read T_i , then use the depth formula.

6.3.3 Limitations and Errors

- **Velocity Inversion (Blind Zone):** If a hard layer overlies a soft layer (e.g., concrete road over sand), waves refract downward away from the surface. The soft layer is never detected.
- **Hidden Layer:** If a layer is very thin, the wave from the layer below might overtake it before it is recorded as a first arrival.
- **Dipping Layers:** If the bedrock is sloped, shooting from one direction (Forward Shot) gives different results than shooting from the other (Reverse Shot). Both are needed to calculate dip.

Past Exam Questions**Frequently Asked Questions:**

- **2072 Magh [6 Marks]:** Explain seismic refraction method of groundwater exploration.